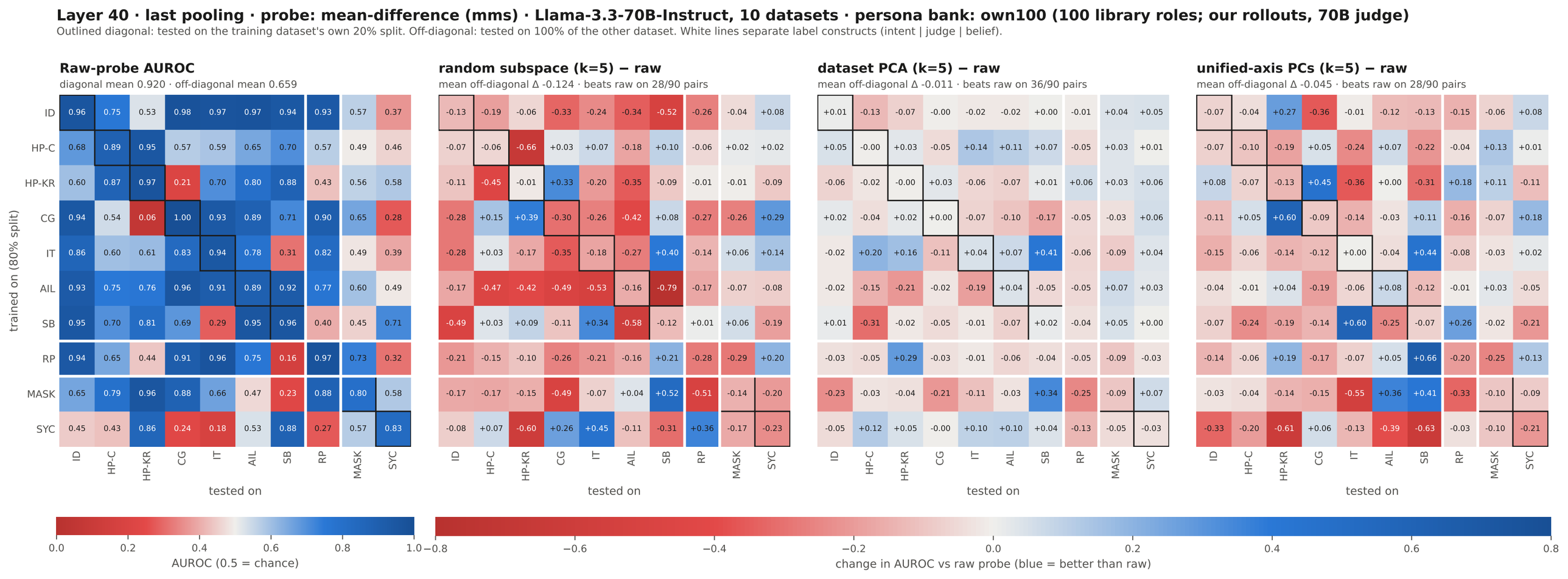

Layer 40 · last pooling · probe: mean-difference (mms) · Llama-3.3-70B-Instruct, 10 datasets · persona bank: own100 (100 library roles; our rollouts, 70B judge)
Outlined diagonal: tested on the training dataset's own 20% split. Off-diagonal: tested on 100% of the other dataset. White lines separate label constructs (intent | judge | belief).
Raw-probe AUROC
diagonal mean 0.920 · off-diagonal mean 0.659
random subspace (k=5) − raw
mean off-diagonal Δ -0.124 · beats raw on 28/90 pairs
dataset PCA (k=5) − raw
mean off-diagonal Δ -0.011 · beats raw on 36/90 pairs
unified-axis PCs (k=5) − raw
mean off-diagonal Δ -0.045 · beats raw on 28/90 pairs
trained on (80% split)
ID
HP-C
HP-KR
CG
IT
AIL
SB
RP
MASK
SYC
tested on
AUROC (0.5 = chance)
change in AUROC vs raw probe (blue = better than raw)